

Wednesday Reading Assessment: Unit 4, Circular Motion

Prof. Jordan C. Hanson

October 15, 2025

1 Memory Bank

- $\Delta s = r\Delta\theta$... Arc length, radius, and angle (radians)
- $\omega = \frac{\Delta\theta}{\Delta t}$... Definition of angular velocity
- $v = r\omega$... Relationship between tangential velocity, v , and angular velocity, ω
- $C = 2\pi r$... Circumference of a circle
- $a_C = r\omega^2$... Centripetal acceleration
- $F_C = mr\omega^2$... Centripetal force

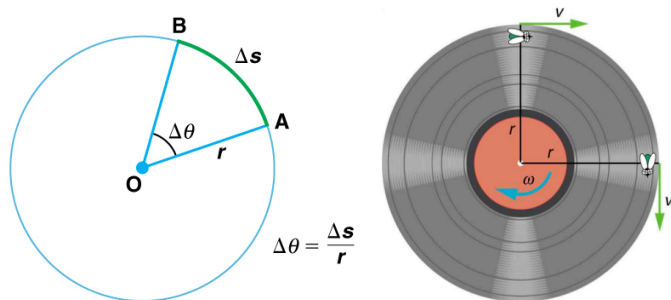


Figure 1: (Left) A circle with a radius r . Arc length Δs around the edge is related to the angle $\Delta\theta$. (Right) A record that is spinning counter-clockwise.

2 Angular Displacement, Velocity

1. Suppose disc has a radius of 0.1 m. (a) What is the circumference? Suppose the object begins to rotate. A spot on the edge returns to its original location after 1.5 seconds. (b) Compute the ratio of the circumference to the time to get a speed.
2. In the previous exercise, how many times per minute will this object rotate?

3. Consider Fig. 1 (right). A record spins at 45 revolutions per minute (rpm). (a) How many revolutions per second is this? (b) Using a formula from the memory bank, calculate the *angular velocity*. Take 2π as $\Delta\theta$, and the time of one rotation as Δt . (c) If the record is left playing for one hour, and continues to spin at 45 rpm, how many times will it rotate?

3 Centripetal Force

1. What is the *centripetal acceleration* of the fly on the edge of the disc in Fig. 1, if the radius is 10 cm?
2. What is the *centripetal force* of the fly, if $m = 1$ gm?

3. Using definitions in the Memory Bank, show that if $F_C = mr\omega^2$, that

$$F_C = \frac{mv^2}{r} \quad (1)$$